TE 458 — Coverage Planning: Tutorial 1 Practice Set (Variant B) — Solutions

4 new numeric variants of Tutorial 1's problems (5G NR downlink/uplink, UMTS indoor, WCDMA macro-cell), same methods, new numbers

> These are freshly parameterised variants of Tutorial 1's four problems — same methods and formulas (see the companion Formula Sheet), different input numbers — for additional self-test practice. Propagation-model coefficients are given directly in each question, exactly as they would be handed to you on an exam.

Problem 1 — 5G NR Downlink at 2.1 GHz

(a) EIRP

$$EIRP = P_{TX} + G_{TX} - L_{TX} = 44 + 17 - 2.5$$

EIRP = 58.5 dBm

(b) Bandwidth

$$\text{Noise floor} = N_0 + 10\log_{10}(W) + NF$$
$$10\log_{10}(W) = -95 - (-173.98) - 7 = 71.98$$
$$W = 10^{7.198}$$

W ≈ 15.8 MHz

(c) Receiver Sensitivity

$$S_i = \text{Noise floor} + E_b/N_0 = -95 + 11$$

S_i = –84 dBm

(d) Maximum Allowable Path Loss (MAPL)

$$MAPL = EIRP - S_i - M_{shadow} - M_{interference}$$
$$MAPL = 58.5 - (-84) - 8 - 4$$

MAPL = 130.5 dB

(e) Cell Radius

$$MAPL = 132.4 + 37.2\log_{10}(R)$$
$$\log_{10}(R) = (130.5 - 132.4) / 37.2 = -0.0511$$

$R = 10^{-0.0511}$ ≈ 0.889 km (≈ 889 m)

Problem 2 — 5G NR Uplink at 2.1 GHz (Log-Distance Model)

(a) EIRP

$$P_{TX} = 10\log_{10}(100) = 20.0\ \text{dBm}$$
$$EIRP = P_{TX} + G_{TX} - L_{TX} = 20.0 + 0 - 0$$

EIRP = 20.0 dBm

(b) Free-Space Path Loss at d_0 = 100 m

$$PL_0 = 20\log_{10}(f) + 20\log_{10}(d) - 147.56$$
$$= 20\log_{10}(2.1\times10^9) + 20\log_{10}(100) - 147.56$$
$$= 186.44 + 40.00 - 147.56$$

PL_0 ≈ 78.88 dB

(c) Noise Calculations at T_0 = 300 K

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i. N₀ = 30 + 10log₁₀(kT₀) = 30 + 10log₁₀(1.38×10⁻²³×300)
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$N_0 \approx -173.83$ dBm/Hz

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ii. Thermal noise power = N₀ + 10log₁₀(B) = −173.83 + 10log₁₀(15×10⁶) = −173.83
+ 71.76
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Thermal noise power ≈ –102.07 dBm

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iii. Noise floor = Thermal noise power + NF = −102.07 + 6.5
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Noise floor ≈ –95.57 dBm

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iv. Receiver sensitivity = Noise floor + required SNR = −95.57 + 15
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Rx sensitivity ≈ –80.57 dBm

(d) Maximum Allowable Path Loss (MAPL)

$$MAPL = EIRP + G_{RX} - L_{RX} - \text{Rx sensitivity} - L_{body} - L_{bldg} - \text{Margin}$$
$$= 20.0 + 15 - 1.5 - (-80.57) - 3 - 8 - 10$$

MAPL ≈ 93.07 dB

(e) Coverage Radius and Area (Log-Distance Model, n = 3.2, d_0 = 100 m)

$$PL(d) = PL(d_0) + 10n \cdot \log_{10}(d/d_0)$$
$$93.07 = 78.88 + 10(3.2) \cdot \log_{10}(d/100) = 78.88 + 32 \cdot \log_{10}(d/100)$$
$$\log_{10}(d/100) = (93.07 - 78.88)/32 = 0.4433$$
$$d = 100 \times 10^{0.4433}$$

i. R ≈ 277.5 m (≈ 0.278 km)

$$Area = (3\sqrt{3}/2)R^2 = 2.598 \times (0.2775)^2$$

ii. Area ≈ 0.200 km²

(f) Effect of a 4 dB Improvement in Receiver Sensitivity

A 4 dB more sensitive receiver tolerates 4 dB more path loss for the same transmit conditions:

$$MAPL_{new} = 93.07 + 4 = 97.07 \text{ dB}$$
$$\log_{10}(d/100) = (97.07 - 78.88)/32 = 0.5683 \Rightarrow d \approx 370.1 \text{ m}$$
$$Area_{new} = 2.598 \times (0.3701)^2 \approx 0.356 \text{ km}^2$$

R_{new} ≈ 0.370 km; coverage area increases by ≈ 77.8% (from ≈ 0.200 km² to ≈ 0.356 km²).

Problem 3 — UMTS Indoor Link Budget (64 kbps, 12 users)

Step 1: Processing Gain

$$G_p = 10\log_{10}(M_{cps}/R_s) = 10\log_{10}(3.84 \times 10^6 / 64 \times 10^3) = 10\log_{10}(60)$$

$G_p \approx 17.78$ dB

Step 2: Receiver Noise Power

$$RNP = N_0 + 10\log_{10}(M_{cps}) + NF = -174 + 10\log_{10}(3.84 \times 10^6) + 6 = -174 + 65.84 + 6$$

RNP ≈ –102.16 dBm

Step 3: Total Noise Interference

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TNI = RNP + Interference margin = −102.16 + 3
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TNI ≈ –99.16 dBm

Step 4: Effective Sensitivity (with N = 12 users)

Since the problem states "considering 12 users", include the $10\log_{10}(N)$ term:
$S_e = 10\log_{10}(N) + TNI - G_p + E_b/N_0$
$= 10\log_{10}(12) + (-99.16) - 17.78 + 3 = 10.79 - 99.16 - 17.78 + 3$

S_e ≈ –103.15 dBm

Step 5: Mobile EIRP

$EIRP = P_{TX} + G_{TX} - L_{body} = 21 + 0 - 0$

EIRP = 21 dBm

(b) Maximum Allowable Path Loss

$MAPL = EIRP - S_e + G_{BS} - L_{feeder} - M_{ff} - M_{sf} - L_{bldg} + G_{SHO}$
$= 21 - (-103.15) + 17 - 2.5 - 5 - 8 - 12 + 2.5$

MAPL ≈ 116.15 dB

(c) Coverage Radius

$MAPL = 137.4 + 35.2 \log_{10}(R)$
$\log_{10}(R) = (116.15 - 137.4)/35.2 = -0.6037$

$R = 10^{-0.6037}$ ≈ 0.249 km (≈ 249 m)

Problem 4 — WCDMA Uplink: 144 kbps Video in Dense Urban

Part (a) Step-by-Step Link Budget

i. $EIRP = P_{TX} + G_{TX} - L_{body} = 24 + 1 - 3$

EIRP = 22 dBm

ii. Receiver Noise Power
$P_N = N_0 + 10\log_{10}(M_{cps}) + NF = -174 + 10\log_{10}(3.84\times10^6) + 5 = -174 + 65.84 + 5$

P_N ≈ –103.16 dBm

iii. Total Noise-Plus-Interference Power
$I_{total} = P_N + MI = -103.16 + 6$

I_{total} ≈ –97.16 dBm

iv. Processing Gain
$G_p = 10\log_{10}(3.84\times10^6/144\times10^3) = 10\log_{10}(26.67)$

G_p ≈ 14.26 dB

v. Effective Receiver Sensitivity (no user count given ⇒ base formula, no $10\log_{10}(N)$ term)
$S_e = I_{total} + E_b/N_0 - G_p = -97.16 + 5 - 14.26$

S_e ≈ –106.42 dBm

vi. Maximum Allowable Path Loss
$MAPL = EIRP - S_e - M_{sf} + G_{SHO} = 22 - (-106.42) - 8 + 2$

MAPL ≈ 122.42 dB

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vii. Cell Radius (given model PLmax = 135.0 + 36.0 log10(R))
log10(R) = (122.42 − 135.0)/36.0 = −0.3494
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R ≈ 0.447 km (≈ 447 m)

Part (b) Quantitative Impact Analysis

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Baseline coverage area: A = 2.598×(0.447)² ≈ 0.519 km²
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(i) Bit Rate Increased to 288 kbps (same E_b/N_0)

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New processing gain: Gp′ = 10log10(3.84×10^6/288×10^3) = 10log10(13.33) ≈ 11.25
dB
New Se′ = −97.16 + 5 − 11.25 ≈ −103.41 dBm
New MAPL′ = 22 − (−103.41) − 8 + 2 ≈ 119.41 dB
log10(R′) = (119.41 − 135.0)/36.0 = −0.4331 ⇒ R′ ≈ 0.369 km
A′ = 2.598×(0.369)² ≈ 0.353 km²
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R′ ≈ 369 m; coverage area shrinks by ≈ 32.0% (from ≈ 0.519 km² to ≈ 0.353 km²).

(ii) Interference Margin Rises to 9 dB (higher network load, bit rate back to 144 kbps)

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New total noise + interference: I″total = −103.16 + 9 = −94.16 dBm
Processing gain unchanged: 14.26 dB
New Se″ = −94.16 + 5 − 14.26 ≈ −103.42 dBm
New MAPL″ = 22 − (−103.42) − 8 + 2 ≈ 119.42 dB
log10(R″) = (119.42 − 135.0)/36.0 = −0.4328 ⇒ R″ ≈ 0.369 km
A″ = 2.598×(0.369)² ≈ 0.354 km²
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R″ ≈ 369 m; coverage area shrinks by ≈ 31.9% (from ≈ 0.519 km² to ≈ 0.354 km²).

Trade-off Discussion

Both changes remove almost exactly the same amount (≈3 dB) from the link budget — one because a higher bit rate needs a smaller processing gain, the other because more interference directly raises the noise floor — and both shrink the coverage radius by ≈17% and the coverage area by ≈32%. This is "cell breathing": loading a CDMA cell with more traffic or higher bit rates shrinks its effective radius, because the added interference (or added bit-rate demand) eats directly into the same link margin that otherwise buys coverage distance. A site sized for coverage at light load can develop coverage holes at its edge once load or throughput demand rises.